# DETECTION PROCESS

UNTIL NOW WE HAVE DEFINED THE FIELDS AND THEIR RELATIONS WITH THE ENVIRONMENT. LET'S NOW COMBINE THE EQUATIONS TO START TO RETRIEVE INFORMATION FROM THE SENSED DOMAIN.

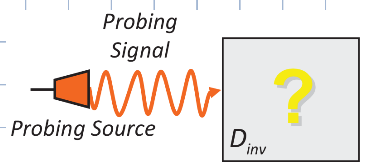
THE EM DETECTION PROCESS CONSISTS OF THE FOLLOWING SEQUENCE:

1) ILLUMINATION
2) MEASUREMENT
3) DATA INVERSION

## ILLUMINATION

A SOURCE $\underline{\tilde{J}}_0(\underline{r})$ RADIATES $\underline{\tilde{E}}_{inc}(\underline{r})$ IN THE REFERENCE SCENARIO

$$\underline{\tilde{E}}_{inc}(\underline{r}) = j\omega\mu \int_{D_{source}} \underline{\tilde{J}}_0(\underline{r}') \cdot \underline{\underline{G}}(\underline{r}|\underline{r}')\, d\underline{r}'$$

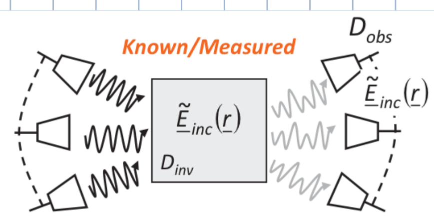



## MEASUREMENT

THE FIELD IS MEASURED THROUGH A PROBE BOTH WITH AND WITHOUT OBJECTS.

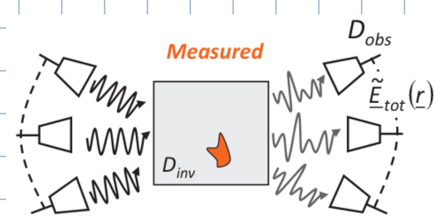
TO DEFINE THE REFERENCE SCENARIO AGAINST WHICH THE FINAL MEASUREMENT IS COMPARED, THE FIELD IS PHYSICALLY MEASURED IN THE OBSERVATION DOMAIN AND IN THE INVESTIGATION DOMAIN WITHOUT ANY OBJECT.

$$\underline{\tilde{E}}_{inc}(\underline{r}) \quad , \quad \underline{r} \in D_{inv} \cup D_{obs}$$



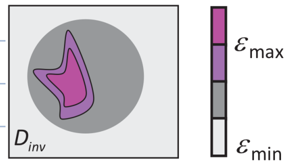

THE FINAL MEASUREMENT IS CARRIED OUT WITH THE OBJECT(S) AND ONLY IN THE OBSERVATION DOMAIN (CAN'T DO IT IN THE INVESTIGATION BECAUSE WOULD NEED TO ENTER INSIDE THE OBJECT → INVASIVE).

$$\underline{\tilde{E}}_{tot}(\underline{r}) \quad , \quad \underline{r} \in D_{obs}$$




## DATA INVERSION

FROM THE PERFORMED MEASUREMENTS, RECONSTRUCT THE OBJECT FUNCTION $\tau(\underline{r})$, $\underline{r} \in D_{inv}$

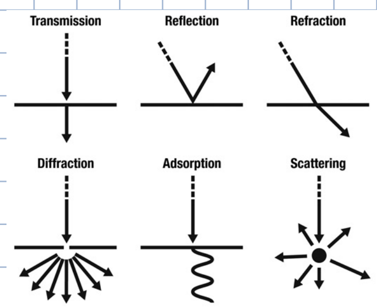


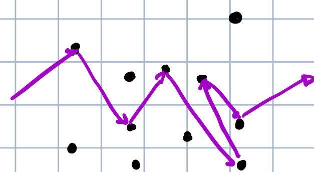

## INVERSE SCATTERING

THE PROCESS OF INVERTING A PROBLEM FROM DATA THAT IS DERIVED FROM THE SCATTERING BETWEEN SOURCE AND OBJECT IS CALLED INVERSE SCATTERING

## SCATTERING

SCATTERING REPRESENTS ANY PROCESS THAT CAUSES A WAVE TO DEVIATE FROM A STRAIGHT TRAJECTORY DUE TO THE INTERACTION WITH THE PROPAGATION MEDIUM.



MULTI-SCATTERING

NOTE THAT THERE EXIST INTERACTIONS THAT ARE NEITHER TRANSMISSION, REFLECTION, ... THESE STILL FALL UNDER THE DEFINITION OF SCATTERING (E.G. MULTI-SCATTERING).

## INVERSE SCATTERING EQUATIONS

TO CARRY ON WITH THE DATA INVERSION, AND THEREFORE EXTRACT INFORMATION ABOUT $\tau(\underline{r})$ IN $D_{INV}$, WE NEED TO DEFINE THE RELATION BETWEEN THE UNKNOWN $\tau(\underline{r})$ AND THE DATA $\longrightarrow$ INVERSE SCATTERING EQUATIONS.

WE KNOW THAT

$$\underline{E}_{scatt}(\underline{r},t) := \underline{E}_{tot}(\underline{r},t) - \underline{E}_{inc}(\underline{r},t)$$

WHERE:

- $\underline{E}_{tot}(\underline{r},t)$ IS MEASURABLE ONLY IN $D_{OBS}$
- $\underline{E}_{inc}(\underline{r},t)$ IS MEASURABLE IN $D_{OBS} \cup D_{INV}$ — TO MEASURE THIS, MUST PERFORM AN ACTUAL MEASUREMENT INSIDE $D_{INV}$
- $\underline{E}_{scatt}(\underline{r},t)$ CAN ONLY BE COMPUTED AND ONLY IN $D_{OBS}$ ($E_{tot}$ IS UNDEFINED ELSEWHERE)

WE CAN THEREFORE SEE THAT THE ONLY MEASURABLE QUANTITIES ARE

$$\underline{E}_{tot}(\underline{r},t) \quad , \quad \underline{r} \in D_{OBS}$$

$$\underline{E}_{inc}(\underline{r},t) \quad , \quad \underline{r} \in D_{OBS}$$

$$\underline{E}_{inc}(\underline{r},t) \quad , \quad \underline{r} \in D_{INV}$$

THIS MEANS THAT, BY COMPUTING THE SCATTERED FIELD JUST FROM THE OBSERVATION DOMAIN, WE ARE ABLE TO OBTAIN INFORMATION ABOUT THE INVESTIGATION DOMAIN WITHOUT GETTING INSIDE IT $\Rightarrow$ NON-INVASIVE!

FOR EACH DATA, LET'S NOW WRITE AN EQUATION WITH THE KNOWN TERMS ON THE LEFT AND THE UNKNOWN ONES ON THE RIGHT:

## 1) DATA EQUATION

$$\tilde{\underline{E}}_{scatt}(\underline{r}) = \overbrace{\tilde{\underline{E}}_{tot}(\underline{r}) - \tilde{\underline{E}}_{inc}(\underline{r})}^{\text{what I measure}} = j\omega\mu \int_{D_{inv}} \tilde{\underline{J}}_{EQ}(\underline{r}')\, \underline{\underline{G}}(\underline{r}|\underline{r}')\, d\underline{r}' \quad , \quad \underline{r} \in D_{OBS}$$

KNOWN COMPUTED — KNOWN MEASURED — UNKNOWN

THE DATA EQUATION REPRESENTS THE NORMAL MEASUREMENT FROM THE SYSTEM'S RECEIVERS. ESSENTIALLY, IT CAN BE THOUGHT OF AS

$$\text{SCATT} = \text{READ} - \text{REFERENCE}$$

## 2) STATE EQUATION

$$\overbrace{\tilde{\underline{E}}_{inc}(\underline{r})}^{\text{what I measure}} = \tilde{\underline{E}}_{tot}(\underline{r}) - \tilde{\underline{E}}_{scatt}(\underline{r}) = \tilde{\underline{E}}_{tot}(\underline{r}) - j\omega\mu \int_{D_{inv}} \tilde{\underline{J}}_{EQ}(\underline{r}')\, \underline{\underline{G}}(\underline{r}|\underline{r}')\, d\underline{r}' \quad , \quad \underline{r} \in D_{inv}$$

KNOWN MEASURED — UNKNOWN IN $D_{INV}$

THE STATE EQUATION REPRESENTS THE CALIBRATION MEASUREMENT. IT IS JUST THE DATA EQUATION "INVERTED", BUT IN THE INVESTIGATION DOMAIN.

THIS INFORMATION IS RETRIEVED BY PERFORMING A MEASUREMENT PHYSICALLY INSIDE THE INVESTIGATION DOMAIN TO MAP THE REFERENCE SCENARIO WITHOUT THE OBJECT (FOR EXAMPLE, IN AN MRI MACHINE, MEASURE THE FIELDS WITH A THIRD ANTENNA WHERE THE PATIENT WOULD LAY).

## PRIMARY / SECONDARY UNKNOWNS

THE PRIMARY UNKNOWN, THE ONE WE ARE REALLY INTERESTED IN, IS THE OBJECT FUNCTION $\tau(\underline{r})$. THE SECONDARY UNKNOWN INSTEAD IS THE TOTAL FIELD IN THE INVESTIGATION DOMAIN. WE ARE INTERESTED IN IT, BUT IT DEPENDS ON $\tau(\underline{r})$, BESIDES DEPENDING ON $\tilde{\underline{J}}_0(\underline{r})$

## DISCRETIZATION

TO SOLVE THE AMIRS PROBLEM, WE NEED TO DETERMINE (BUILD AN IMAGE OF) THE PRIMARY UNKNOWN $\tau(\underline{r})$ AND THE SECONDARY UNKNOWN $\tilde{\underline{E}}_{tot}(\underline{r})$ FROM JUST THE STATE AND DATA EQUATIONS.

TO DO THAT, WE MUST FIRST RECOGNIZE THAT THE TWO EQUATIONS ARE OF PARTICULAR TYPE THAT ALLOWS US TO USE A STATE OF THE ART METHOD.

## CLASSIFICATION OF INTEGRAL EQUATIONS

FOR SIMPLICITY, LET'S CONSIDER SCALAR EQUATIONS (VECTOR EQUATIONS WOULD SIMPLY BE 3 SCALAR EQUATIONS) OF THE FOLLOWING FORM:

$$s(x) = \alpha(x) f(x) + \int_{a(x)}^{b(x)} f(x')\, g(x|x')\, dx'$$

WHERE:

- $s(x)$, $\alpha(x)$, $g(x|x')$ ARE KNOWN FUNCTIONS
- $f(x)$ IS THE UNKNOWN FUNCTION
- $g(x|x') = g(x-x')$ IS THE KERNEL OPERATOR (SOME CONCEPT OF GREEN'S FUNCTION)

INTEGRAL EQUATIONS CAN BE CLASSIFIED BASED ON:

1) THE INTEGRATION EXTREMES $a(x)$, $b(x)$

$$s(x) = \alpha(x) f(x) + \int_{a(x)}^{b(x)} f(x')\, g(x|x')\, dx'$$

- IF $a(x)$ AND $b(x)$ ARE CONSTANTS, THEY ARE CALLED FREDHOLM'S EQUATIONS

$$\begin{cases} a(x) = a \\ b(x) = b \end{cases}$$

- IF INSTEAD $a(x)$ AND $b(x)$ ARE FUNCTIONS OF $x$, THEY ARE CALLED VOLTERRA'S EQUATIONS

$$\begin{cases} a(x) = \xi(x) \\ b(x) = \psi(x) \end{cases}$$

2) THE FUNCTION $\alpha(x)$

$$s(x) = \alpha(x) f(x) + \int_{a(x)}^{b(x)} f(x')\, g(x|x')\, dx'$$

- IF $\alpha(x) = 0$, THE EQUATION IS OF THE FIRST KIND

$$s(x) = \int_{a(x)}^{b(x)} f(x')\, g(x|x')\, dx'$$

- IF $\alpha(x) = \alpha$, THE EQUATION IS OF THE SECOND KIND

$$s(x) = \alpha f(x) + \int_{a(x)}^{b(x)} f(x')\, g(x|x')\, dx'$$

- IF $\alpha(x) = \Psi(x)$, THE EQUATION IS OF THE THIRD KIND

$$s(x) = \alpha(x) f(x) + \int_{a(x)}^{b(x)} f(x')\, g(x|x')\, dx'$$

NOTE THAT THESE CONCEPTS ARE DIRECTLY EXTENDED TO VECTOR EQUATIONS.

IF WE NOW GO BACK TO THE STATE AND DATA EQUATIONS WE CAN SEE THAT

1) THE STATE EQUATION IS A FREDHOLM'S VECTOR EQUATION OF THE SECOND KIND

$$\underline{\tilde{E}}_{INC}(\underline{r}) = \underline{\tilde{E}}_{TOT}(\underline{r}) - j\omega\mu \int_{D_{INV}} \underline{\tilde{J}}_{EQ}(\underline{r}')\, \underline{\underline{G}}(\underline{r}|\underline{r}')\, d\underline{r}'$$

$\underline{s}(\underline{r})$ $\qquad$ $\alpha(x) \neq 0$ $\underline{f}(\underline{r})$ $\qquad$ $\underline{\underline{g}}(\underline{r}, \underline{r}')$ (KERNEL)

2) THE DATA EQUATION IS A FREDHOLM'S VECTOR EQUATION OF THE FIRST KIND

$$\underbrace{\tilde{\underline{E}}_{scatt}(\underline{r})}_{s(\underline{r})} = \underbrace{j\omega\mu}_{\substack{\alpha(x)=0 \\ \text{CONSTANT}}} \int_{D_{INV}} \underbrace{\tilde{\underline{J}}_{EQ}(\underline{r}')}_{f(\underline{r})} \underbrace{\underline{\underline{G}}(\underline{r}|\underline{r}')}_{\underline{\underline{g}}(\underline{r},\underline{r}')\ (\text{KERNEL})} d\underline{r}'$$

## ANALYTICAL SOLUTION

CAN WE SOLVE THESE EQUATIONS ANALYTICALLY?

AN ANALYTICAL SOLUTION IMPLIES IT SUPPORTS A CLOSED FORM EXPRESSION. IN SIMPLER WORDS, THE SOLUTION MUST BE EXPRESSED AS A FUNCTION OF THE DATA THROUGH A FINITE NUMBER OF MATHEMATICAL OPERATIONS.

$$\sum_{n=1}^{\infty} \text{ IS NOT A CLOSED FORM}$$

BESIDE CANONICAL OBJECTS (CYLINDER, SPHERE, ...) AND IDEAL SOURCES (PLANE WAVES), IT IS NOT POSSIBLE TO SOLVE FREDHOLM'S EQUATIONS ANALYTICALLY / IN CLOSED FORM.

## DISCRETIZATION OF ANALYTICAL EQUATIONS

TO SOLVE THE PROBLEM, WE RECUR TO NUMERICAL SOLUTIONS, WHICH APPROXIMATE THE EXACT SOLUTION BY DISCRETIZING THE DESCRIPTIVE EQUATIONS.

WHAT WE WANT IS A METHOD OF SOLVING EQUATIONS OF THE IMPLICIT FORM

$$L\{\underbrace{f(x)}_{\text{UNKNOWN}}\} = \underbrace{s(x)}_{\text{KNOWN}}$$

WHERE $L\{\}$ IS AN INTEGRO-DIFFERENTIAL OPERATOR

$$L\{f(x)\} = \alpha(x) f(x) + \int_a^b f(x') g(x,x') dx'$$

WHICH WE RECOGNIZE TO BE A FREDHOLM'S EQUATION.

AMONG THE MANY METHODS, WE CONSIDER THE METHOD OF MOMENTS (MoM), THAT EXPANDS THE FUNCTION $f(x)$ IN A LINEAR COMBINATION OF N KNOWN BASIS FUNCTIONS. IT WORKS ESSENTIALLY LIKE THE FOURIER TRANSFORM, WHERE THE BASIS FUNCTIONS WOULD BE COSINES, OR LIKE THE WAVELET TRANSFORM.

$$f(x) = \sum_{n=1}^{N} f_n q_n(x)$$

WHERE:
- $q_n(x)$ IS THE n-TH BASIS FUNCTION
- $f_n$ IS THE n-TH COEFFICIENT

THIS GENERIC DEFINITION ALLOWS US TO CHOOSE ANY SET OF BASIS FUNCTIONS, ALLOWING US TO USE THE BEST ONE FOR THE SPECIFIC APPLICATION.

IN OUR SPECIFIC CASE, WE GET

$$s(x) = \alpha(x) f(x) + \int_a^b f(x') g(x,x') dx' \longrightarrow s(x) = \sum_{n=1}^{N} f_n \alpha(x) q_n(x) + \sum_{n=1}^{N} f_n \int_a^b q_n(x') g(x,x') dx'$$

$$\underset{\text{KNOWN}}{S(x)} = \underbrace{\sum_{n=1}^{N} f_n \left[ \alpha(x) q_n(x) + \int_a^b q_n(x')\, g(x,x')\, dx' \right]}_{\text{UNKNOWN}}$$

THE GOAL NOW IS TO FIND THE BEST BASIS FUNCTIONS AND, TO DO SO, THERE IS A MATHEMATICAL APPROACH.
FIRST, WE MUST CONSIDER AN ADDITIONAL SET OF FUNCTIONS: TEST/WEIGHT FUNCTIONS

$$\{\psi_m \,;\; m = 1, \dots, M\}$$

THESE ARE CHOSEN BY THE USER DEPENDING ON THE APPLICATION. ONCE CHOSEN, WE PROJECT OUR KNOWN FUNCTION $S(x)$ ON EACH TEST FUNCTION VIA THE DOT PRODUCT, DEFINING THE MOMENT OF $S(x)$

$$\langle S(x), \psi_m(x) \rangle = \int_a^b S(x)\, \psi_m(x)\, dx$$

WHICH CAN BE EXPANDED TO

$$\int_a^b S(x)\, \psi_m(x)\, dx = \int_a^b \left\{ \sum_{n=1}^{N} f_n \left[ \alpha(x) q_n(x) + \int_a^b q_n(x')\, g(x,x')\, dx' \right] \right\} \psi_m(x)\, dx$$

$$\underbrace{\int_a^b S(x)\, \psi_m(x)\, dx}_{S_m} = \sum_{n=1}^{N} f_n \left\{ \underbrace{\int_a^b \alpha(x) q_n(x) \psi_m(x)\, dx}_{r_{mn}} + \underbrace{\int_a^b \psi_m(x) \left[ \int_a^b q_n(x')\, g(x,x')\, dx' \right] dx}_{g_{mn}} \right\}, \quad m = 1, \dots, M$$

WHICH WE CAN REWRITE IN COMPACT FORM AS

$$\underset{\text{KNOWN}}{S_m} = \sum_{n=1}^{N} \underset{\text{UNKNOWN}}{f_n} \underbrace{\left( r_{mn} + g_{mn} \right)}_{\text{KNOWN}}, \quad m = 1, \dots, M$$

WHERE

$$S_m = \int_a^b S(x)\, \psi_m(x)\, dx$$

$$r_{mn} = \int_a^b \alpha(x) q_n(x) \psi_m(x)\, dx \qquad \text{(KNOWN COEFFICIENT)}$$

$$g_{mn} = \int_a^b \psi_m(x) \left[ \int_a^b q_n(x')\, g(x,x')\, dx' \right] dx$$

THE COMPACT FORM CAN BE FURTHER COMPACTED IN MATRIX FORM

$$\underline{S} = [R]\, \underline{f}$$

$$\begin{bmatrix} S_1 \\ S_2 \\ \vdots \\ S_M \end{bmatrix} = \begin{bmatrix} r_{11}+g_{11} & & \\ & \ddots & \\ & & r_{MN}+g_{MN} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_M \end{bmatrix}$$

WHAT WE ARE MISSING NOW IS A METHOD FOR CHOOSING THE BASIS AND TEST FUNCTIONS. WHEN CHOOSING THEM, OUR GOALS ARE: SIMPLIFICATION, GOOD APPROXIMATION AND GOOD MATRIX CONDITIONING.

a) SIMPLIFICATION: WE WANT THE KNOWN MATRICIAL COEFFICIENTS TO BE OF CLOSED/ANALYTICAL FORM, I.E. CHOOSE FUNCTIONS THAT SIMPLIFY THE MATRIX $[R]$

b) GOOD APPROXIMATION: WE WANT TO ACHIEVE THE BEST APPROXIMATION OF THE UNKNOWN FUNCTION $f(x)$ WHILE REDUCING THE NUMBER OF UNKNOWN EXPANSION COEFFICIENTS $N$.

THIS BRINGS THE FOLLOWING ADVANTAGES:

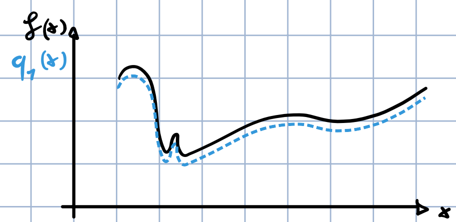
- THE MATRIX $[R]$ HAS REDUCED SIZE
- THE NUMBER OF NECESSARY DATA $(M)$ IS REDUCED

IF MATRIX IS SQUARE, CAN SIMPLY INVERT PROBLEM AND COMPUTE SOLUTION

$$[R] \in \mathbb{C}^{M \times N} \longrightarrow \text{IF } M = N, \quad \underline{f} = [R]^{-1}\, \underline{S}$$

IN THE BEST-CASE SCENARIO A SINGLE BASIS FUNCTION IS SUFFICIENT: IF WE KNOW EXACTLY HOW $f(x)$ BEHAVES, WE CAN CHOOSE THAT EXACT FUNCTION AS BASIS FUNCTION $\longrightarrow$ NEED ONLY ONE COEFFICIENT.

IN ANOTHER CASE, IF THE BASIS FUNCTIONS ARE VERY SIMILAR TO THE UNKNOWN FUNCTION $f(x)$, WE NEED VERY FEW COEFFICIENTS.




$$f(x) = \sum_{n=1}^{N} f_n\, q_n(x) \longrightarrow f(x) = f_1\, q_1(x)$$

c) GOOD MATRIX CONDITIONING: WE WANT $[R]$ TO BE A WELL-CONDITIONED MATRIX TO ENSURE AN EASIER MATRIX INVERSION

$$R \cdot R^* \text{ WILL HAVE EIGENVALUES} \longrightarrow \text{CONDITION NUMBER} = \max\{\text{EIGENVALUES}\} - \min\{\text{EIGENVALUES}\}$$

LET'S NOW SEE TWO EXAMPLE METHODS FOR CHOOSING TEST FUNCTIONS: POINT-MATCHING MoM AND GALERKIN MoM.

## POINT-MATCHING MoM

WITH PM-MoM, THE WEIGHTING FUNCTIONS ARE A SET OF M DELTAS CENTERED IN DIFFERENT POINTS $x_m$

$$\Psi_m(x) = \delta(x - x_m), \quad m = 1, \dots, M$$

WHICH ALLOWS US TO USE THE SAMPLING THEOREM

$$S_m = \int^{b} S(x)\, \Psi_m(x)\, dx \longrightarrow S_m = \int^{b} S(x)\, \delta(x - x_m)\, dx = s(x_m)$$

$$v_{mn} = \int_a^b \alpha(x) q_n(x) \psi_m(x)\, dx \longrightarrow v_{mn} = \int_a^b \alpha(x) q_n(x) \delta(x - x_m)\, dx = \alpha(x_m) q_m(x_m)$$

$$g_{mn} = \int_a^b \psi_m(x) \left[ \int_a^b q_n(x') g(x,x')\, dx' \right] dx \longrightarrow g_{mn} = \int_a^b \delta(x - x_m) \left[ \int_a^b q_n(x') g(x,x')\, dx' \right] dx = \int_a^b q_n(x') g(x_m, x')\, dx'$$

WITH THIS, THE COMPLETE DISCRETIZED EQUATION BECOMES

$$s(x) = \sum_{n=1}^{N} f_n \left[ \alpha(x) q_n(x) + \int_a^b q_n(x') g(x,x')\, dx' \right] \longrightarrow s(x_m) = \sum_{n=1}^{N} f_n \left[ \alpha(x_m) q_n(x_m) + \int_a^b q_n(x') g(x_m,x')\, dx' \right]$$

WHICH IS THE ORIGINAL CONTINUOUS EQUATION SAMPLED IN $x_m$, WHICH IS THE POINT IN WHICH THE DATA IS MEASURED (I.E. WHERE THE PROBE IS PLACED)

## GALERKIN MOM

THE TEST FUNCTIONS ARE CHOSEN TO BE THE SAME (BUT CONJUGATE) AS THE BASIS FUNCTIONS, BOTH IN THE SAME DOMAIN OF $f(x)$.

$$f(x) : D \to R \quad \Rightarrow \quad q_n(x) : D \to R \quad , \quad \psi_m(x) : D \to R$$

$$\psi_m(x) = q_n^*(x)$$

# DISCRETIZATION OF THE DATA EQUATION

NOW THAT WE HAVE THE METHOD, LET'S DISCRETIZE THE DATA EQUATION WITH PM-MOM. UP UNTIL NOW WE HAVE ANALYZED THE DISCRETIZATION OF SCALAR EQUATIONS, BUT THE DATA EQUATION IS VECTORIAL. WE CAN SIMPLY WORK ON EACH COMPONENT SEPARATELY!

$$\underline{\tilde{E}}_{scatt}(\underline{r}) = j\omega\mu \int_{D_{INV}} \underline{\tilde{J}}_{EQ}(\underline{r}') \underline{\underline{G}}(\underline{r}|\underline{r}')\, d\underline{v}'$$

$$\downarrow$$

$$\tilde{E}_q^{scatt}(\underline{r}) = j\omega\mu \int_{D_{INV}} \left[ \underline{\tilde{J}}_{EQ}(\underline{r}') \underline{\underline{G}}(\underline{r}|\underline{r}') \right] \cdot \hat{\underline{q}}\, d\underline{v}'$$

PROJECTED ON THE COMPONENT $q$

$$\tilde{E}_q^{scatt}(\underline{r}) = \underline{\tilde{E}}_{scatt}(\underline{r}) \cdot \hat{\underline{q}} \quad , \quad q \in \{x, y, z\}$$

LOOK AT SEPARATE COMPONENTS INDIVIDUALLY

OBSERVE THAT

DYADIC : $\underline{\underline{D}} = \sum_i \underline{D}_i \hat{\underline{i}} = \sum_i \underbrace{\sum_p D_{p_i} \hat{\underline{p}}}_{\underline{D}_i} \hat{\underline{i}}$

SUM OF VECTORS, EACH FOR A DIRECTION

$\hat{\underline{i}} \cdot \hat{\underline{i}} = 1$

$$\underline{v} \cdot \underline{\underline{D}} = \left( \sum_i v_i \hat{\underline{i}} \right) \cdot \left( \sum \sum D_{p_i} \hat{\underline{p}} \hat{\underline{i}} \right) = \sum v_i \sum D_{p_i} \hat{\underline{p}}$$

VECTOR: $\underline{v} = \sum_i v_i \, \hat{i}$

SINGLE COMPONENTS

SINCE WE WANT TO SCALARIZE THE FUNCTION, WE NEED TO SEPARATE IT INTO ITS COMPONENTS BY PROJECTING IT ON SOME DIRECTION $\hat{q} = \{\hat{x}, \hat{y}, \hat{z}\}$

$$\left(\underline{v} \cdot \underline{\underline{D}}\right) \cdot \hat{q} = \left(\sum_i v_i \sum_p D_{p_i} \, \hat{p}\right) \cdot \hat{q}$$

AND SINCE $\hat{p} = \{\hat{x}, \hat{y}, \hat{z}\}$ AND $\hat{q} = \{\hat{x}, \hat{y}, \hat{z}\}$, $\hat{p} \cdot \hat{q} \neq 0 \Leftrightarrow \hat{p} = \hat{q}$, OTHERWISE $\hat{p} \cdot \hat{q} = 1$. THEREFORE

ORTHOGONAL BASIS

$$\left(\underline{v} \cdot \underline{\underline{D}}\right) \cdot \hat{q} = \sum_i v_i D_{q_i}$$

IF WE NOW SUBSTITUTE $\underline{v}$ AND $\underline{\underline{D}}$ WITH $\underline{J_{EQ}}$ AND $\underline{\underline{G}}$, WE OBTAIN

$$\left(\underline{J_{EQ}} \cdot \underline{\underline{G}}\right) \cdot \hat{q} = \sum_i J_i^{EQ} \sum_p G_{p_i} \, \hat{p} \cdot \hat{q} = \sum_i J_i^{EQ} G_{q_i}$$

SINCE THEY SHARE THE SAME FORM, THE DATA EQUATION SPLIT INTO INDIVIDUAL COMPONENTS BECOMES

$$\tilde{E}_q^{scatt}(\underline{r}) = j\omega\mu \sum_{i=1}^{3} \int_{D_{INV}} \tilde{J}_i^{EQ}(\underline{r}') \, G_{qi}(\underline{r}|\underline{r}') \, d\underline{v}' \quad , \quad q = x, y, z$$

OBSERVATION: IF WE CONSIDER AN ISOTROPIC OBJECT, THE OBJECT FUNCTION IS THE SAME IN ALL DIRECTIONS AND DOES NOT NEED TO BE SEPARATED INTO ITS COMPONENTS.

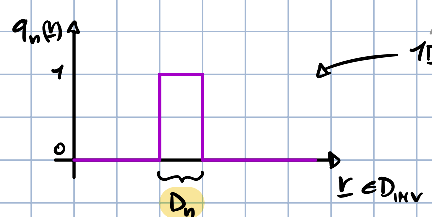
$$\tilde{J}_i^{EQ}(\underline{r}) = \tau(\underline{r}) \, \tilde{E}_i^{tot}(\underline{r})$$

NOW THAT WE MANAGED TO RECONDUCT OUR VECTORIAL PROBLEM TO 2D, WE NEED TO DISCRETIZE.
WE START BY EXPANDING THE UNKNOWN FUNCTION $f(x) \longrightarrow \tilde{J}_i(\underline{r})$ IN BASIS FUNCTIONS

$$f(x) = \sum_{n=1}^{N} f_n \, q_n(x) \longrightarrow \tilde{J}_i(\underline{r}) = \sum_{n=1}^{N} \left(\tilde{J}_i\right)_n q_n(\underline{r})$$

WHERE THE BASIS FUNCTION $q_n(\underline{r})$ IS CHOSEN TO BE A RECTANGULAR IMPULSE ON THE DOMAIN

$$q_n(\underline{r}) = \begin{cases} 1 & \underline{r} \in D_n \big|_{\underline{r} \in D_{INV}} \\ 0 & \underline{r} \notin D_n \big|_{\underline{r} \in D_{INV}} \end{cases}$$




ONE SINGLE COMPONENT/DIRECTION

1D REPRESENTATION, IN 3D IT WOULD BE A CUBE

WE CAN NOW PLUG IN THE DATA EQUATION AND REPRESENT IT AS A SUM OF BASIS FUNCTIONS

$$E_q^{scatt}(\underline{r}) = j\omega\mu \sum_{i=1} \int_{D_{INV}} \sum_{n=1} (\tilde{J}_i^{EQ})_n \, q_n(\underline{r}) \, G_{qi}(\underline{r}|\underline{r}') \, d\underline{v}' \quad , \quad q = x,y,z$$

NOTE THAT THIS EQUATION IS $\neq 0$ ONLY INSIDE $D_n$, BECAUSE EVERYWHERE ELSE $q_n(\underline{r}) = 0$. WHAT THIS IS DOING IS ESSENTIALLY RESTRICTING THE INTEGRATION DOMAIN TO ONLY $D_n$.
IN FACT, WE CAN REWRITE IT AS

$$\tilde{E}_q^{scatt}(\underline{r}) = j\omega\mu \sum_{i=1}^{3} \sum_{n=1}^{N} \underbrace{(\tilde{J}_i^{EQ})_n}_{\text{CURRENT DENSITY IN DIRECTION } i \text{ IN THE } n\text{-TH CUBE}} \int_{D_n} G_{qi}(\underline{r}|\underline{r}') \, d\underline{v}' \quad , \quad q = x,y,z$$

$D_n \to$ n-TH SUB-DOMAIN CUBE
$\Downarrow$
INTEGRATE $\forall$ CUBE

RECALL THE SAMPLING THEOREM:

$$\int_x f(x) \, \delta(x - x_m) \, dx = f(x_m)$$

IF WE APPLY IT TO OUR EQUATION WE GET

$$E_q^{scatt}(\underline{r_m}) = j\omega\mu \sum_{i=1}^{3} \sum_{n=1}^{N} (\tilde{J}_i^{EQ})_n \int_{D_n} G_{qi}(\underline{r_m}|\underline{r}') \, d\underline{v}' \quad , \quad q = x,y,z$$

$G_{mn}$

WHICH WE CAN COMPACT BY DEFINING A NEW MATRIX $G_{mn}$ OR $\{G_{qi}\}_{mn}$

$$E_q^{scatt}(\underline{r_m}) = \sum_{i=1}^{3} \sum_{n=1}^{N} (\tilde{J}_i^{EQ})_n \, G_{mn} \quad , \quad q = x,y,z$$

$$\boxed{G_{mn} = \{G_{qi}\}_{mn} = j\omega\mu \int_{D_n} G_{qi}(\underline{r_m}, \underline{r}') \, d\underline{v}'}$$

WE CAN REWRITE THE EQUATION IN MATRIX FORM AS FOLLOWS

EXTERIOR

$$\underbrace{[E^{scatt}]}_{3M} = \underbrace{[G^{ext}]}_{3M \times 3N} \underbrace{[J]}_{3N}$$

| | | |
|---|---|---|
| Where | $[E^{scatt}] \in C^{3M\times 1}$ | $[E^{scatt}] = \{[E_q^{scatt}]; \; q = x,y,z\}^T$ |
| | $[E_q^{scatt}] \in C^{M\times 1}$ | $[E_q^{scatt}] = \{\tilde{E}_q^{scatt}(\underline{r}_m); \; m = 1,\ldots,M\}^T$ |
| | $[J] \in C^{3N\times 1}$ | $[J] = \{[J_q]; \; q = x,y,z\}^T$ |
| | $[J_q] \in C^{N\times 1}$ | $[J_q^{scatt}] = \{(\tilde{J}_q)_n; \; n = 1,\ldots,N\}^T$ |
| | $[G^{ext}] \in C^{3M\times 3N}$ | $[G^{ext}] = \{[G^{ext}]_{qi}; \; q,i = x,y,z\}$ |
| | $[G^{ext}]_{qi} \in C^{M\times N}$ | $[G^{ext}]_{qi} = \{(G_{qi}^{ext})_{mn}; \; m = 1,\ldots,M; \; n = 1,\ldots,N\}^T$ |

M = # OF OBSERVATION POINTS (TEST FUNCTIONS)
N = # OF DOMAIN DISCRETIZATION BLOCKS

THE SAME PROCESS CAN BE DONE TO DISCRETIZE J:

$$\underline{\tilde{J}_{EQ}}(\underline{r}) = \underline{\underline{\tau}}(\underline{r}) \cdot \underline{\tilde{E}_{TOT}}(\underline{r})$$

$\hookrightarrow$ ANISOTROPIC MEDIUM

AS DONE BEFORE, WE PROJECT ON $\hat{q}$ AND APPLY THE BASIS FUNCTION EXPANSION:

$$\tau_{pq}(\underline{r}) = \sum_{n=1}^{N} (\tau_{pq})_n \, q_n(\underline{r}) \qquad \tilde{E}_q^{tot}(\underline{r}) = \sum_{n=1}^{N} (E_q^{tot})_n \, q_n(\underline{r})$$

WHICH, COMBINED WITH THE DATA EQUATION, RESULTS IN

$$E_q^{scatt}(\underline{r_m}) = \{(\tau_{pq})_n (E_q^{tot})_n\}\{G_{qi}\}_{mn} \quad , \quad q = x, y, z \quad , \quad m = 1, \dots, M \quad , \quad \underline{r_m} \in D_{OBS}$$

$$\downarrow$$

$$\underbrace{[E^{scatt}]}_{3M} = \underbrace{[G^{ext}]}_{3M \times 3N} \underbrace{[\tau]}_{3N \times 3N} \underbrace{[E^{tot}]}_{3N}$$

where

$[\tau] \in C^{3N \times 3N}$ $\quad [\tau] = \{[\tau_{pq}];\ p,q = x,y,z\}^T$

$$[\tau_{pq}] \in C^{N \times N} \quad [\tau_{pq}] = \begin{bmatrix} (\tau_{pq})_1 \ddots & 0 & 0 \\ 0 & (\tau_{pq})_n \ddots & 0 \\ 0 & 0 & (\tau_{pq})_N \end{bmatrix} \qquad [\tau_{pq}] = \begin{cases} [\tau_{pq}]_{tn} = (\tau_{pq}) & if \ t = n \\ [\tau_{pq}]_{tn} = 0 & if \ t \neq n \\ t, n = 1, \dots, N \end{cases}$$

$[E^{tot}] \in C^{3N \times 1}$ $\quad [E^{tot}] = \{[E_q^{tot}];\ q = x,y,z\}^T$

$[E_q^{tot}] \in C^{N \times 1}$

$[E_q^{tot}] = \{\tilde{E}_q^{tot}(\underline{r}_n);\ n = 1, \dots, N\}^T$

IN THE PARTICULAR CASE OF AN ISOTROPIC MEDIUM, WE HAVE THAT

IDENTITY DYADIC

$$\underline{\tilde{J}_{EQ}}(\underline{r}) = \underbrace{\tau(\underline{r})\,\underline{\underline{I}}}\cdot \underline{\tilde{E}_{tot}}(\underline{r}) = \tau(\underline{r})\,\underline{\tilde{E}_{tot}}(\underline{r})$$

$$[\tau_{pq}] = \begin{cases} \sum_{n=1}^{N} \tau_n q_n(\underline{r}) & , \ p = q \\ 0 & , \ p \neq q \end{cases}$$

AND THE DISCRETIZED DATA EQUATION BECOMES

$$E_q^{scatt}(\underline{r_m}) = \sum_{i=1}^{3} \sum_{i=1}^{N} \{\tau_n (\tilde{E}_q^{tot})_n\}\{G_{qi}\}_{mn}$$

$$\downarrow$$

$$[E^{scatt}] = [G^{ext}][\tau][E^{tot}]$$

$$[\tau] = \{[\tau_{pq}];\ p,q = x,y,z\}^T \qquad [\tau_{pq}] = \begin{cases} [\tau_{pq}]_{tn} = (\tau_{pq}) & if \ t = n \\ [\tau_{pq}]_{tn} = 0 & if \ t \neq n \\ t, n = 1, \dots, N \end{cases}$$

$$\boxed{[\tau_{pq}] = \begin{bmatrix} \tau_1 & 0 & 0 \\ 0 & \tau_n & 0 \\ 0 & 0 & \tau_N \end{bmatrix}}$$

## DISCRETIZATION OF THE STATE EQUATION

SIMILARLY TO THE DATA EQUATION, WE NOW WANT TO DISCRETIZE THE STATE EQUATION WITH PM-MOM.
AS DONE BEFORE, WE EXPAND THE VECTORIAL EQUATION INTO IT'S SINGLE COMPONENTS.

$$\underline{\tilde{E}_{INC}}(\underline{r}) = \underline{\tilde{E}_{TOT}}(\underline{r}) - j\omega\mu \int_{D_{INV}} \underline{\tilde{J}_{EQ}}(\underline{r}')\,\underline{\underline{G}}(\underline{r}|\underline{r}')\,d\underline{r}' \quad , \underline{r} \in D_{INV}$$

$$\downarrow$$

$$\tilde{E}_q^{INC}(\underline{r}) = \tilde{E}_q^{TOT}(\underline{r}) - j\omega\mu \sum_{i=1}^{3} \int_{D_{INV}} \tilde{J}_i(\underline{r}')\,G_{qi}(\underline{r}|\underline{r}')\,d\underline{r}' \quad , \underline{r} \in D_{INV}$$

THEN, WE EXPAND THE 2 UNKNOWNS $\tilde{E}_q^{TOT}$ AND $\tilde{J}_i$ WITH BASIS FUNCTIONS

$$\tilde{E}_q^{TOT}(\underline{r}) = \sum_{n=1}^{N} \left(\tilde{E}_q^{TOT}\right)_n q_n(\underline{r}) \qquad\qquad \tilde{J}_i(\underline{r}) = \sum_{n=1}^{N} \left(\tilde{J}_i\right)_n q_n(\underline{r})$$

DON'T DEPEND ON POSITION, INFORMATION IS EMBEDDED IN BASIS FUNCTIONS.
JUST COEFFICIENTS → ASSUME CONSTANT WHERE $q_n = 1$

BEING $q_n(\underline{r})$ RECTANGULAR IMPULSES, WE GET

$$\tilde{E}_q^{INC}(\underline{r}) = \sum_{n=1}^{N} \left(\tilde{E}_q^{TOT}\right)_n q_n(\underline{r}) - j\omega\mu \sum_{i=1}^{3} \sum_{n=1}^{N} \left(\tilde{J}_i\right)_n \int_{D_n} G_{qi}(\underline{r}|\underline{r}')\,d\underline{r}' \quad , \underline{r} \in D_{INV} , \ q = x, y, z$$

NOW, WE DEFINE M TEST FUNCTIONS (IN $D_{INV}$ NOW, NOT $D_{OBS}$ !)

$$\Psi_m(\underline{r}) = \delta(\underline{r} - \underline{r_m}) \quad , \ \underline{r_m} \in D_{INV} \quad , \ m = 1, \dots, M$$

SAMPLING THE INVESTIGATION DOMAIN (OBSERVATION POINTS)

TO COMPUTE THE MOMENT OF THE DATA

$$\left\langle \tilde{E}_q^{INC}(\underline{r}), \Psi_m(\underline{r}) \right\rangle = \int_{D_{INV}} \tilde{E}_q^{INC}(\underline{r})\,\delta(\underline{r} - \underline{r_m})$$

$$= \sum_{n=1}^{N} \left(\tilde{E}_q^{TOT}\right)_n \int_{D_{INV}} q_n(\underline{r_m})\,\delta(\underline{r} - \underline{r_m})\,d\underline{r} - j\omega\mu \sum_{i=1}^{3} \sum_{n=1}^{N} \left(\tilde{J}_i\right)_n \int_{D_{INV}} \int_{D_n} G_{qi}(\underline{r_m}|\underline{r}')\,\delta(\underline{r} - \underline{r_m})\,d\underline{r}'d\underline{r}$$

BY APPLYING THE SAMPLING THEOREM, WE THEN GET

$$\left\langle \tilde{E}_q^{INC}(\underline{r}), \delta(\underline{r} - \underline{r_m}) \right\rangle = \tilde{E}_q^{INC}(\underline{r_m}) = \sum_{n=1}^{N} \left(\tilde{E}_q^{TOT}\right)_n q_n(\underline{r_m}) - j\omega\mu \sum_{i=1}^{3} \sum_{n=1}^{N} \left(\tilde{J}_i\right)_n \int_{D_n} G_{qi}(\underline{r_m}|\underline{r}')\,d\underline{r}'$$

SINCE $q_n(\underline{r_m})$ IS A RECTANGLE, IT IS ≠0 ONLY IF $\underline{r_m}$ IS INSIDE THAT TRIANGLE, I.E. $n = m$

$$\tilde{E}_q^{INC}(\underline{r_m}) = \left(\tilde{E}_q^{TOT}\right)_m - j\omega\mu \sum_{i=1}^{3} \sum_{n=1}^{N} \left(\tilde{J}_i\right)_n \int_{D_n} G_{qi}(\underline{r_m}|\underline{r}')\,d\underline{r}'$$

IN COMPACT FORM THIS BECOMES

$$\tilde{E}_q^{inc}(\underline{r}_\mu) = \left(\tilde{E}_q^{tot}\right)_\mu - \sum_{i=1}^{3}\sum_{n=1}^{N}\left(\tilde{J}_i\right)_n \{G_{qi}\}_{\mu n}$$

WHERE

$$\{G_{qi}\}_{\mu n} = j\omega\mu\int_{D_n} G_{qi}(\underline{r}_\mu|\underline{r}')\,d\underline{r}'$$

IN MATRIX FORM IT BECOMES

$$\underset{3M}{\left[E^{inc}\right]} = \underset{3M}{\left[E^{tot}\right]} - \underset{3M\times 3N}{\left[G^{int}\right]}\underset{3N}{\left[J\right]}$$

Where

$$\left[E^{inc}\right] \in C^{3U\times 1} \qquad \left[E^{inc}\right] = \left\{\left[E_q^{inc}\right];\ q = x,y,z\right\}^T$$

$$\left[E_q^{inc}\right] \in C^{U\times 1} \qquad \left[E_q^{inc}\right] = \left\{\tilde{E}_q^{inc}\left(\underline{r}_u\right);\ u = 1,\ldots,U\right\}^T$$

$$\left[J\right] \in C^{3N\times 1} \qquad \left[J\right] = \left\{\left[J_q\right];\ q = x,y,z\right\}^T$$

$$\left[J_q\right] \in C^{N\times 1} \qquad \left[J_q\right] = \left\{\left(\tilde{J}_q\right)_n;\ n = 1,\ldots,N\right\}^T$$

$$\left[G^{int}\right] \in C^{3U\times 3N} \qquad \left[G^{int}\right] = \left\{\left[G^{int}\right]_{qi};\ q,i = x,y,z\right\}$$

$$\left[G^{int}\right]_{qi} \in C^{U\times N} \qquad \left[G^{int}\right]_{qi} = \left\{\left(G_{qi}^{int}\right)_{un};\ u = 1,\ldots,U;\ n = 1,\ldots,N\right\}^T$$

IN THE GENERAL CASE OF AN ANISOTROPIC MEDIUM

$$\underline{\tilde{J}_{EQ}}(\underline{r}) = \underline{\underline{\tau}}(\underline{r})\cdot\underline{\tilde{E}_{tot}}(\underline{r})$$

AND THE EQUATION BECOMES

$$\underset{3M}{\left[E^{inc}\right]} = \underset{3M}{\left[E^{tot}\right]} - \underset{3M\times 3N}{\left[G^{int}\right]}\underset{3N\times N}{\left[\tau\right]}\underset{3N}{\left[E^{tot}\right]}$$

Where

$$\left[\tau\right] \in C^{3N\times N}$$

$$\left[E^{tot}\right] \in C^{3N\times 1}$$

# SOLUTION

THE EM-ANIRS PROBLEM FORMULATED BY MEANS OF THE DATA AND STATE EQUATIONS IS:

- ILL-POSED: THE SOLUTION EXISTS, BUT IT IS NOT UNIQUE AND DOES NOT HAVE CONTINUOUS DEPENDENCE ON DATA

REQUIRES OPTIMIZATION


- NON-LINEAR: THE DATA DEPENDS ON THE PRODUCT $J_{EQ} \propto \tau E_{TOT}$

THE SOLUTION OF THE EM-ANIRS PROBLEM IS DEFINED AS

$$\tau_{OPT} = \underset{\tau}{\mathrm{argmin}} \{ \Phi(\tau) \}$$

OPTIMUM

WHERE $\Phi$ IS A COST FUNCTION THAT QUANTIFIES THE MISMATCH BETWEEN THE OBTAINED SOLUTION AND THE TRUE ONE IN AN APPROPRIATE FUNCTIONAL SPACE.

THERE ARE MULTIPLE COST FUNCTIONS, EACH WITH ITS OWN RESOLUTION TECHNIQUE. A COST FUNCTION IS CONSIDERED BAD IF

- AN ALGORITHM CAN'T FIND THE MINIMUM OF $\Phi$
- $\Phi$ IS NOT DEFINED WELL AND, EVEN IF $\tau = \tau_{REAL}$, $\Phi \neq 0$

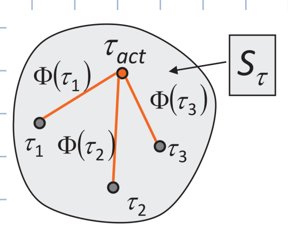
TO MINIMIZE THE COST FUNCTION WE THEREFORE NEED BOTH A SUITABLE FUNCTION AND AN APPROPRIATE ALGORITHM

BEFORE DEFINING THEM, LET'S RECALL THE DEFINITION OF VECTOR NORM

$$\|\underline{v}\|_p = \sqrt[p]{\sum_{n=1}^{N} |v_n|^p}$$

## IDEAL COST FUNCTION

IF THE CHOSEN FUNCTIONAL SPACE IS THE SPACE OF OBJECT FUNCTIONS $S_\tau$, THE COST FUNCTION IS THE DISTANCE BETWEEN THE CURRENT AND THE ACTUAL $\tau$




IF $\int_{D_{INV}} |\tau_{ACT}(\underline{r})|^2 d\underline{r} = 0$ CONSIDER ONLY NUMERATOR

$$\Phi_{IDEAL}(\tau) = \sqrt{\frac{\int_{D_{INV}} |\tau(\underline{r}) - \tau_{ACT}(\underline{r})|^2 d\underline{r}}{\int_{D_{INV}} |\tau_{ACT}(\underline{r})|^2 d\underline{r}}} \xrightarrow[\substack{\tau(\underline{r}) \to \tau(\underline{r}_n) \\ n=1,\dots,N}]{\text{DISCRETIZATION}} \Phi_{IDEAL}(\tau) \simeq \sqrt{\frac{\sum_n |\tau(\underline{r}_n) - \tau_{ACT}(\underline{r}_n)|^2}{\sum_n |\tau_{ACT}(\underline{r}_n)|^2}} = \frac{\|\tau - \tau_{ACT}\|_2}{\|\tau_{ACT}\|_2}$$

NOTE THAT THE IDEAL COST FUNCTION COINCIDES WITH THE RECONSTRUCTION ERROR AND THEREFORE THE CONDITION

$$\Phi_{IDEAL}(\underline{\tau}) = 0 \iff \underline{\tau} = \tau_{ACT}$$

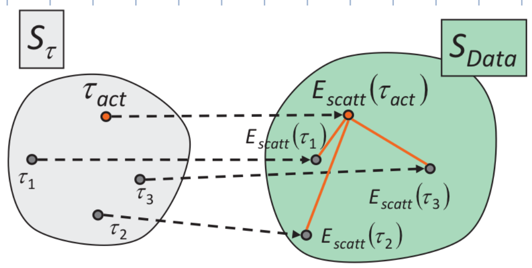
IS SUFFICIENT FOR THE OPTIMALITY OF THE SOLUTION

THIS DEFINITION REQUIRES PRIOR KNOWLEDGE OF $\tau_{ACT}$, WHICH IS NOT REALISTIC → ONLY IDEAL, NOT DOABLE.
TO AVOID THIS LIMIT, WE CONSIDER COST FUNCTIONS IN ALTERNATIVE SPACES, SUCH AS THE SPACE OF DATA, OF THE STATE OR BOTH.

## COST FUNCTION IN THE SPACE OF DATA

THE IDEA IS THAT WE WANT TO KEEP THE STRUCTURE OF THE IDEAL COST FUNCTION, BUT REPLACE $\tau_{ACT}$ WITH SOMETHING COMPUTABLE/MEASURABLE WHICH IS AS SIMILAR/CORRELATED TO $\tau_{ACT}$ AS POSSIBLE: SOMETHING THAT IS A FUNCTION OF $\tau_{ACT}$.

CONSIDER THE SPACE OF DATA $S_{DATA}$ (A PROJECTED SPACE W.R.T. $S_\tau$) → I.E. THE KNOWLEDGE OF $E_{TOT}/E_{SCATT}$




$\underline{r_m} \in D_{OBS}$

SINCE WE KNOW THAT $E_{SCATT}$ IS A FUNCTION OF $\tau$, WE CAN DEFINE THE COST FUNCTION ON $E_{SCATT}$ AS THE DISTANCE BETWEEN THE DATA OF THE CURRENT SOLUTION FROM THE ACTUAL ONE.

$$\Phi_{DATA}(\tau) = \frac{\sum_{m=1}^{M} \| \underline{\tilde{E}}_{SCATT}(\underline{r_m}) - \underline{\tilde{E}}^{REC}_{SCATT}(\underline{r_m}) \|_2}{\sum_{m=1}^{M} \| \underline{\tilde{E}}_{SCATT}(\underline{r_m}) \|_2}$$

RECONSTRUCTED

WHERE $\underline{\tilde{E}}^{REC}_{SCATT}(\underline{r}) = j\omega\mu \int_{D_{INV}} \underline{\tilde{J}}_{EQ}(\underline{r}') \cdot \underline{\underline{G}}(\underline{r}|\underline{r}')\, d\underline{r}'$ BEING $\underline{\tilde{J}}_{EQ}(\underline{r}) = \tau(\underline{r})\, \underline{\tilde{E}}_{TOT}(\underline{r})$

OBSERVE THAT:

a) THE COST FUNCTION CAN BE SPLIT INTO ITS x, y, z COMPONENTS

$$\Phi_{DATA}(\tau) = \frac{\sum_m \left[ \sum_q \left| \tilde{E}_q^{SCATT}(\underline{r_m}) - \tilde{E}_q^{SCATT,REC}(\underline{r_m}) \right|^2 \right]}{\sum_m \left[ \sum_q \left| \tilde{E}_q^{SCATT}(\underline{r_m}) \right|^2 \right]}$$

RECONSTRUCTED

b) THE CONDITION $\Phi_{DATA}(\tau) = 0$ IS NECESSARY BUT NOT SUFFICIENT

$$\Phi_{DATA}(\tau) = 0 \implies \underline{\tilde{E}}_{SCATT}(\underline{r_m}) = \underline{\tilde{E}}^{ACT}_{SCATT}(\underline{r_m})$$

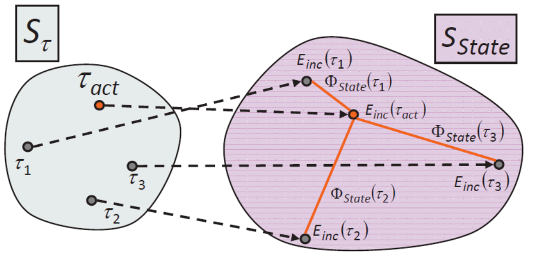
$$\Phi_{DATA}(\tau) = 0 \;\not\Longrightarrow\; \tau(\underline{r_m}) = \tau_{ACT}(\underline{r_m})$$

- SINCE $E_{SCATT} = \tau E_{TOT}$, $E_{TOT}$ COULD BE 0 WHILE $\tau \neq 0$
- DIFFERENT OBJECTS CAN GENERATE THE SAME $E_{SCATT}$

c) EVEN IF $\tau(\underline{r}) = 0$ (OBJECT IS ESTIMATED AS NOT PRESENT), THEN $\Phi_{DATA}(\underline{r}) \neq 0$ IF $\underline{\tilde{E}}_{SCATT}(\underline{r}) \neq 0$ WHEN $\tau_{ACT}(\underline{r}) \neq 0$ (OBJECT IS ACTUALLY PRESENT) ↳ WTFFF?!?!

## COST FUNCTION IN THE SPACE OF STATE

CONSIDER THE SPACE OF STATE $S_{STATE}$ $\longrightarrow$ I.E. THE KNOWLEDGE OF $E_{INC}$




$\underline{r}_\mu \in D_{INV}$

THE COST FUNCTION CAN BE DEFINED AS DONE BEFORE

MEASURED — RECONSTRUCTED

$$\Phi_{STATE}(\tau) = \frac{\sum_{\mu=1}^{U} \| \underline{\tilde{E}}_{INC}(\underline{r}_\mu) - \underline{\tilde{E}}^{REC}_{INC}(\underline{r}_\mu) \|_2}{\sum_{\mu=1}^{U} \| \underline{\tilde{E}}_{INC}(\underline{r}_\mu) \|_2}$$

OBSERVE THAT $\Phi_{STATE} = 0$ IN TWO CONDITIONS, NOT ONLY ONE AS IN $\Phi_{DATA}$

$$\Phi_{STATE} = 0 \iff \underline{\tilde{E}}_{INC}(\underline{r}_\mu) = \underline{\tilde{E}}^{REC}_{INC}(\underline{r}_\mu)$$

$\tau(\underline{r}) = \tau_{ACT}(\underline{r})$

OBJECT = TRUE

$\downarrow$

$E^{MEAS}_{INC} = E^{REC}_{INC}$

$\tau(\underline{r}) = 0$

OBJECT = FREE-SPACE

$\downarrow$

SINCE $E_{INC} = E_{TOT} - j\omega\mu \int \tau \ldots$

IF $\tau = 0 \implies E_{INC} = E_{TOT} \implies E_{SCATT} = 0$

## COST FUNCTION WITH NON-ENERGETIC CONSTRAINTS

THE SECOND CONDITION OF $\Phi_{DATA}$ IS UNDESIRED BY ITSELF, BUT WE CAN COMBINE $\Phi_{STATE}(\tau)$ AND $\Phi_{DATA}(\tau)$ TO AUGMENT THE INFORMATION AVAILABLE. IN PARTICULAR, WE USE $\Phi_{STATE}$ AS A CONSTRAINT SUBJECT TO $\Phi_{DATA}$.

$$\Phi(\tau) = \Phi_{DATA}(\tau) + \alpha\, \Phi_{STATE}(\tau)$$

LAGRANGE MULTIPLIER — NON-ENERGETIC CONSTRAINT